

CANONICAL LESSON

Shelter Before Disaster

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C54
Notebook	V7_D_N04
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Plan emergency shelter by connecting hazards, exposure, vulnerability, accessibility, capacity, routes, uncertainty, safeguarding, and authorized emergency action.

Learning outcomes

Why this matters

A shelter count alone can fail when locations are unsafe, inaccessible, undersized, or disconnected from transport and protection services.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

Worked cross-sector example

An emergency team tests flood scenarios against population needs, accessible routes, shelter capacity, and backup options.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Add route/travel-time constraints.
- 2.
- Model shelter failure.
- 3.

- Add uncertainty in population estimates.
- 4.
- Explain why the highest exposure zone may not receive the first intervention.

Answer key

- 1.
- Use authoritative networks, hazards, modes, barriers, congestion, and accessibility.
- 2.
- Stress-test closure/capacity loss and identify redundancy.
- 3. propagate census/grid uncertainty through scenario totals and rank stability.
- 4.
- Urgency also depends on warning time, feasibility, vulnerability, access, cascading hazards, current evidence, and authorized response strategy.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.